Dayananda Sagar University

School of Engineering

Devarakaggalahalli, Harohalli, Kanakapura Road, Bangalore South Dt., Bengaluru – 562112

Department of Computer Science & Technology

ACADEMIC YEAR 2025-26

Internship Report

(22CT4802)

on

Data Analytics Intern

Submitted in partial fulfilment for the award of degree of

B.Tech in

Computer Science & Technology

Submitted By

YAMUNA V - ENG22CT0052

(8th Semester)

Department of Computer Science and Technology
Dayananda Sagar University

Dayananda Sagar University

School of Engineering

Devarakaggalahalli, Harohalli, Kanakapura Road, Bangalore South Dt., Bengaluru – 562112

Department of Computer Science & Technology

CERTIFICATE

This is to certify that the Internship work done at **" CODTECH IT SOLUTIONS PRIVATE LIMITED "** is being submitted by **YAMUNA V - ENG22CT0052** is the record of the internship carried out by her under our supervision. This report is submitted towards the partial fulfillment of B.Tech in Computer Science & Technology during the academic year **2025-2026**. It is certified that all the suggestions or corrections indicated for internal assessment have been incorporated in the report. This Internship Report has been approved as it satisfies the academic requirements under the rules prescribed for the Bachelor of Technology degree.

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DECLARATION

I, Ms. YAMUNA V (ENG22CT0052) student of eighth semester B.Tech in Computer Science & Technology, at School of Engineering, Dayananda Sagar University, hereby declare that the Internship entitled "Full Stack Web Development Intern" is Bonafide work carried out at CODTECH IT SOLUTIONS PRIVATE LIMITED, is a partial fulfillment for the award of degree in Bachelor of Technology in Computer Science & Technology during the academic year 2025-2026.

YAMUNA V - ENG22CT0052 Student Signature

Place : Bangalore Date:

ACKNOWLEDGEMENT

I am content and grateful for having been provided with this internship opportunity at CODTECH IT SOLUTIONS PRIVATE LIMITED. I take this opportunity to express my deepest gratitude and appreciation to all those who have been helping me directly or indirectly towards the successful completion of this internship.

I would like to thank **Dr. M Shahina Parveen, Chairperson of the Department of Computer Science & Technology,** for her warm support and for providing the required resources to land this internship opportunity.

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I would like to thank **Neela Santhosh Kumar (Human Resources & Academic Head)** for giving me an opportunity to be a part of this project and intern at their company, **CODTECH IT SOLUTIONS PRIVATE LIMITED .**

I am grateful to the staff of the Department of Computer Science & Technology for imparting invaluable knowledge and guiding me throughout the duration of my course as a Computer Science & Technology student at Dayananda Sagar University.

YAMUNA V
B.Tech (CST),
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ABSTRACT

This report presents a detailed overview of the knowledge and practical experience gained during the internship at Codtech IT Solutions Pvt. Ltd. The internship primarily focused on Data Analysis, Machine Learning, and Data Visualization techniques, aiming to bridge the gap between theoretical learning and real-world applications.

During the internship, four major tasks were completed: Exploratory Data Analysis (EDA), House Price Prediction using Machine Learning, Sales Dashboard Development, and Sentiment Analysis of textual data. Each task was designed to enhance understanding of data preprocessing, feature engineering, predictive modeling, and interactive visualization.

The internship provided hands-on experience in using tools and technologies such as Python, Pandas, NumPy, Matplotlib, Scikit-learn, Plotly, and Dash. Through these tasks, real-world datasets were analyzed to extract meaningful insights and build efficient models.

In addition to technical knowledge, the internship also improved problem-solving skills, logical thinking, and the ability to interpret and present data effectively. Overall, this internship significantly contributed to strengthening both technical and analytical skills required in the field of data science.

TABLE OF CONTENTS

CHAPTER 1: COMPANY PROFILE

1. ABOUT THE COMPANY ……... **2**
2. SERVICES AND PRODUCTS …… **2**

CHAPTER 2: SYSTEM REQUIREMENTS

1. HARDWARE REQUIREMENTS …… **3**
2. SOFTWARE REQUIREMENTS …….. **3**

CHAPTER 3: PROJECT WORK

1. PROBLEM STATEMENT …….. **3**
2. TECHNOLOGIES USED …… **3**
3. TRAINING …….. **4**
4. RESULT …… **4 - 11**

CONCLUSION …….... 11 - 12

REFERENCES …… 12

INTERNSHIP OFFER LETTER …….. 13

COMPLETION CERTIFICATE

CHAPTER 1: COMPANY PROFILE

1.1 ABOUT THE COMPANY

Codtech IT Solutions Pvt. Ltd is a technology-driven organization that focuses on providing IT services, software development solutions, and industry-oriented training programs. The company aims to bridge the gap between academic knowledge and industry requirements by offering practical exposure through internships and real-time projects.

The organization specializes in areas such as Data Science, Machine Learning, Web Development, and Software Engineering. It emphasizes hands-on learning, enabling students to gain practical experience by working on real-world datasets and projects.
Codtech promotes a learning environment where students can develop technical skills, understand industry workflows, and improve their problem-solving abilities. The company also focuses on modern tools, frameworks, and best practices such as version control, clean coding, and project documentation.

1.2 SERVICES AND PRODUCTS

Codtech IT Solutions Pvt. Ltd provides a wide range of services and solutions aimed at enhancing technical skills and delivering efficient software solutions.

Services Offered

- Data Analysis and Machine Learning training
- Full Stack Web Development
- Internship programs with real-time projects
- Technical mentoring and project guidance
- Debugging, testing, and deployment support

Products and Solutions

- Data-driven analytical tools
- Machine learning-based prediction models
- Interactive dashboards and visualization systems
- Real-time web applications
- Automation and data processing systems

CHAPTER 2: SYSTEM REQUIREMENTS

2.1 HARDWARE REQUIREMENTS

- Laptop/Desktop with minimum 4GB RAM
- Processor: Intel i3 or above
- Internet connectivity

2.2 SOFTWARE REQUIREMENTS

- Python (3.x)
- Jupyter Notebook / VS Code
- Libraries:
    - Pandas
    - NumPy
    - Matplotlib
    - Scikit-learn
    - Plotly
    - Dash

CHAPTER 3: SYSTEM REQUIREMENTS

3.1 PROBLEM STATEMENT

The primary objective of this internship is to analyze datasets, extract meaningful insights, build predictive models, and develop interactive visualization systems. The tasks aim to demonstrate the application of data science techniques in solving real-world problems.

3.2 TECHNOLOGIES USED

The following tools and technologies were used during the internship:

Programming Language

- Python

Libraries

- Pandas – data manipulation and analysis
- NumPy – numerical computations
- Matplotlib – data visualization
- Seaborn – advanced visualization
- Scikit-learn – machine learning models
- Plotly – interactive visualizations
- Dash – dashboard development

Tools

- VS Code / Jupyter Notebook – coding environment
- GitHub – version control and project submission

3.3 TRAINING

During the internship, structured training was provided to understand both theoretical and practical aspects of data analysis and machine learning.

The training included:

- Understanding data preprocessing techniques
- Learning how to clean and prepare datasets
- Applying statistical methods for data analysis
- Building machine learning models for prediction
- Developing dashboards using visualization tools
- Performing sentiment analysis using NLP techniques

Through continuous practice and task-based learning, hands-on experience was gained in handling real datasets, debugging code, and improving model performance.

3.4 RESULTS

TASK 1: EXPLORATORY DATA ANALYSIS (EDA)

Objective

To understand the structure and characteristics of the dataset using statistical analysis and visualization techniques.

Methodology

- Imported dataset using Pandas
- Analyzed dataset using .info() and .describe()
- Handled missing values and cleaned data
- Performed univariate and bivariate analysis
- Visualized data using histograms, bar charts, and scatter plots

Insights

- Identified patterns and trends in data
- Understood relationships between variables
- Detected outliers and anomalies

Result

EDA helped in gaining a clear understanding of the dataset, which was essential for further model building and analysis.

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.barplot(x="COUNTRY", y="SALES", data=pdf)
plt.xticks(rotation=90)
plt.show()
```

[10]

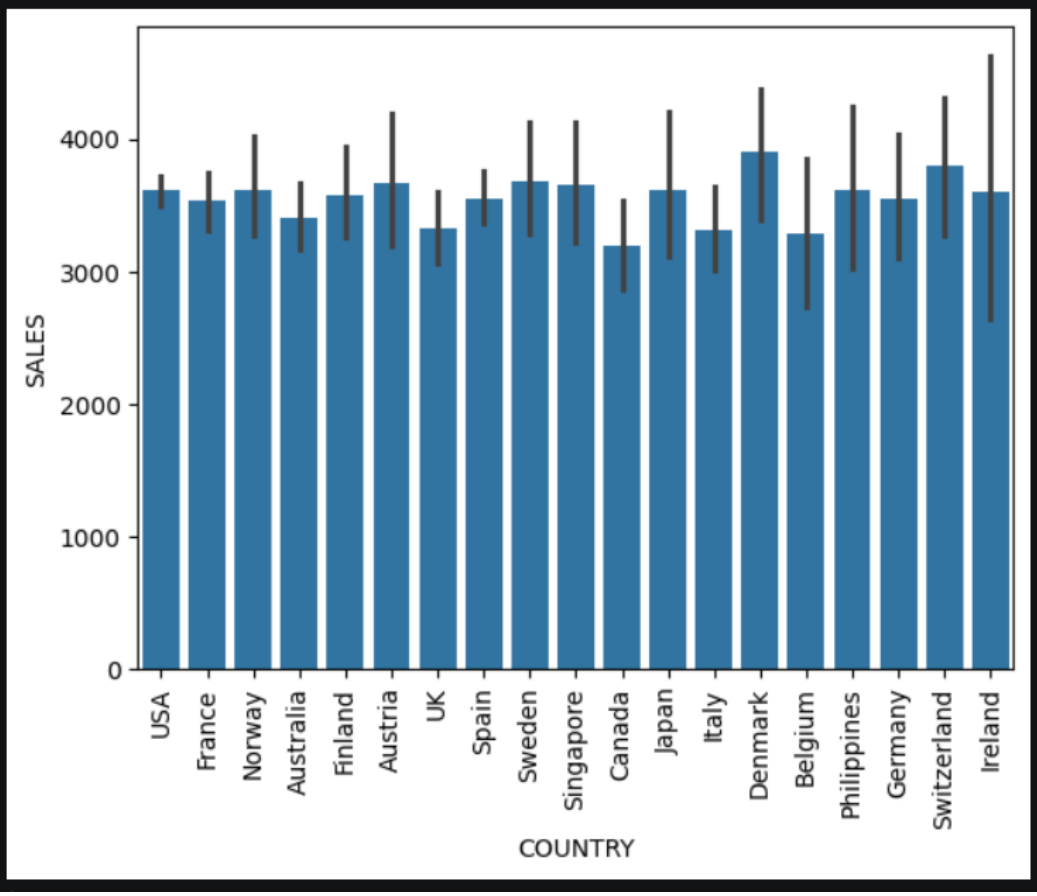

Fig 1: Sales Distribution Across Different Countries

This bar chart shows sales across different countries. It helps compare performance and identify which countries have higher or lower sales.

TASK 2: HOUSE PRICE PREDICTION

Objective

To build a machine learning model that predicts house prices based on various features.

Methodology

- Data preprocessing (handling missing values, encoding categorical data)
- Feature selection
- Splitting dataset into training and testing sets
- Applied regression algorithms
- Evaluated model performance

Model Used

- Linear Regression / Machine Learning Regression Model

Results

- Successfully predicted house prices
- Model showed good accuracy and performance

Conclusion

This task demonstrated how machine learning can be used to solve real-world prediction problems.

```
mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)
[9]
...
Mean Squared Error: 24.291119474973538
R2 Score: 0.6687594935356317
```

Fig 2: Model Evaluation Metrics (MSE and R^2 Score)

This output shows the performance of the regression model using Mean Squared Error (MSE) and R^2 score. The MSE indicates the prediction error, while the R^2 score represents how well the model explains the data.

```
plt.scatter(y_test, predictions)
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices")
plt.show()
[10]
...
```

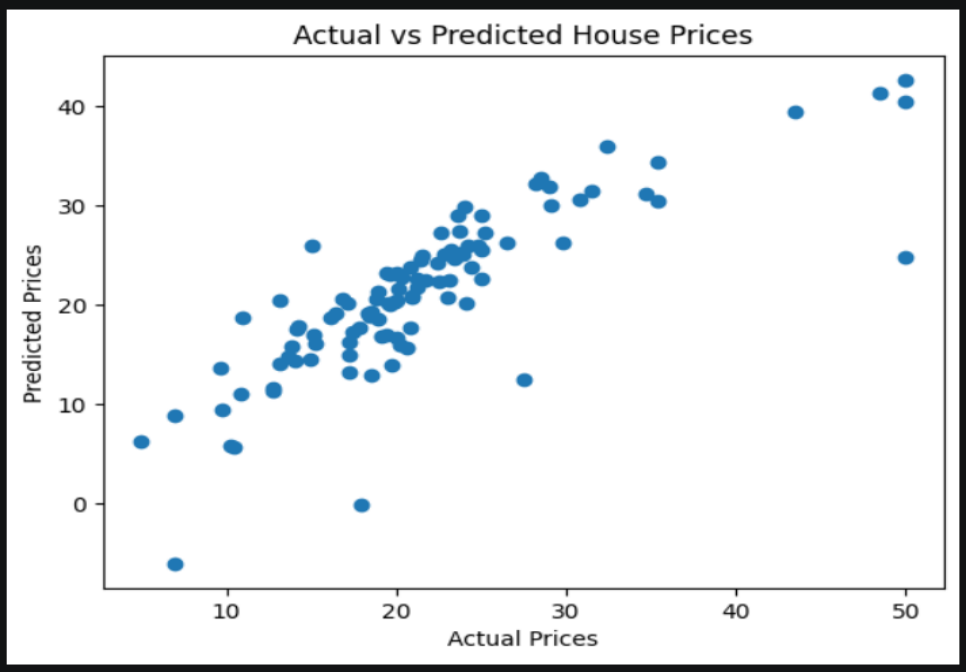

Fig 3: Actual vs Predicted House Prices

This scatter plot compares actual house prices with predicted values from the model. The closer the points are to a straight line, the better the model's predictions, indicating good accuracy.

TASK 3: SALES DASHBOARD

Objective

To develop an interactive dashboard that visualizes sales data.

Implementation

- Used Dash framework for web application
- Used Plotly for visualization
- Created interactive components:
  - Dropdown filter (Country selection)
  - Bar chart (Sales by product line)
  - Line chart (Sales over time)
  - Pie chart (Sales distribution)

Features

- Dynamic updates based on user input
- Clean and user-friendly interface
- Real-time data visualization

Result

The dashboard successfully displayed sales insights and trends, making data easy to understand and analyze.

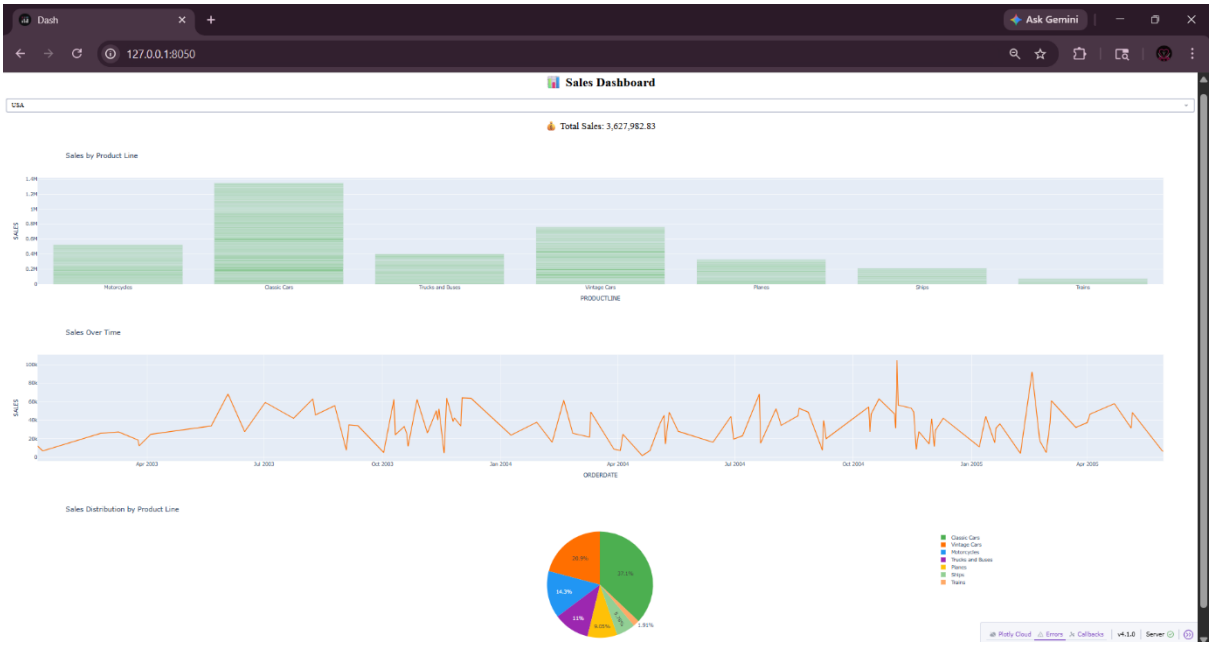

Fig 4: Interactive Sales Dashboard

This dashboard displays sales insights using interactive visualizations, including bar, line, and pie charts. It allows users to analyze sales by product line, track sales over time, and view distribution across categories.

TASK 4: SENTIMENT ANALYSIS

Objective

To classify text reviews into positive and negative sentiments.

Methodology

- Data cleaning and preprocessing
- Converted text to numerical form using TF-IDF vectorization
- Applied Logistic Regression model
- Evaluated model using accuracy and confusion matrix

Results

- Accuracy achieved: **~91%**
- Balanced dataset improved model performance
- Successfully classified new reviews

Insights

- Model relies on keywords like "good", "bad", "excellent"
- Struggles with sarcasm and complex context

```
#MODEL TRAINING
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train_vec, y_train)

#PREDICTION
y_pred = model.predict(X_test_vec)

#EVALUATION
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.91424
```

Fig 5: Logistic Regression Model Training and Accuracy

This output shows the training and evaluation of the Logistic Regression model for sentiment analysis. The model is trained on text data and achieves an accuracy of approximately 91%, indicating good performance in classification.

```
train_df["Sentiment"].value_counts().plot(
    kind="bar",
    color=colors
)

plt.title("Sentiment Distribution", fontsize=14)
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```

[42]

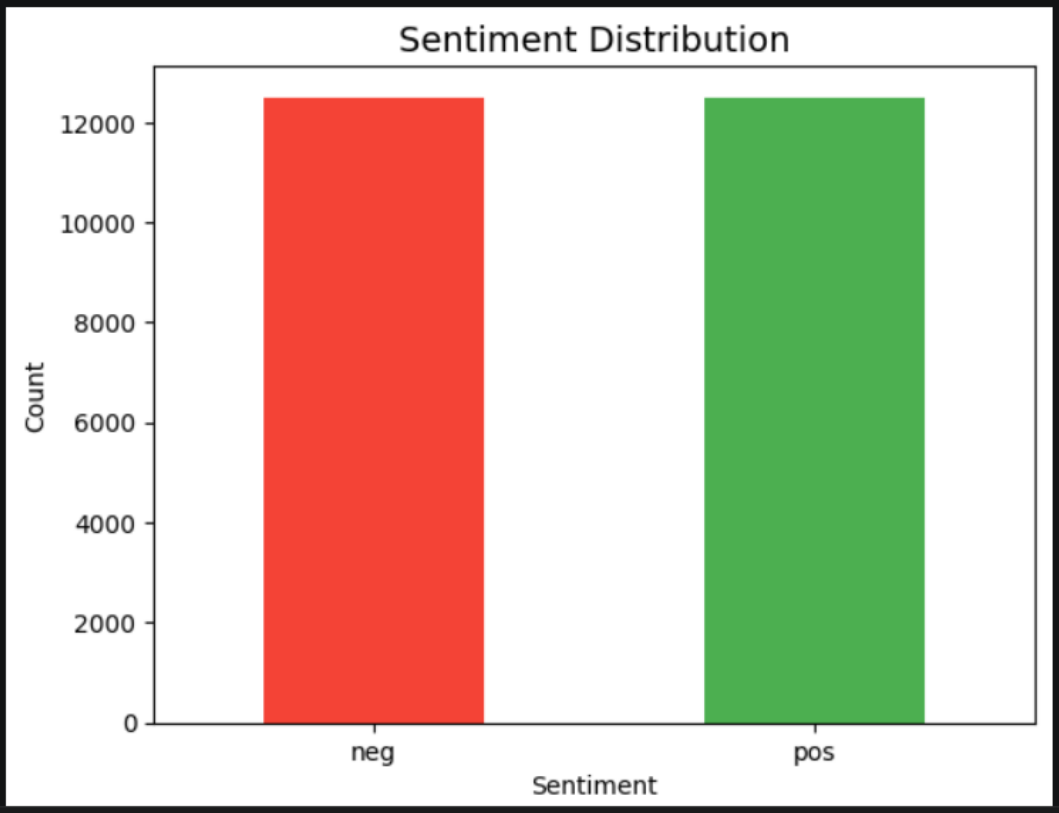

Fig 6: Sentiment Distribution Visualization

This visualization represents the distribution of sentiment labels (positive and negative) in the dataset. The bar chart shows that both sentiment classes are nearly balanced, which helps in reducing bias during model training. A balanced dataset improves the performance and reliability of the sentiment analysis model.

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap="Blues")  # blue theme

plt.title("Confusion Matrix")
plt.show()
```

[45]

...

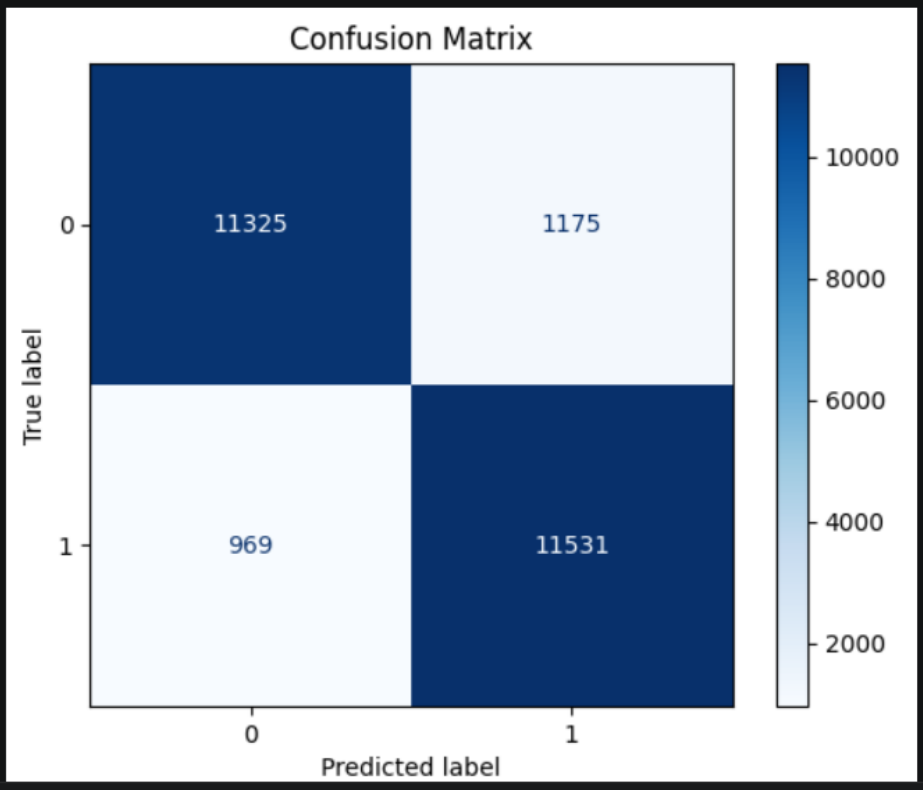

Fig 7: Confusion Matrix

This confusion matrix visualizes the performance of the sentiment analysis model by comparing actual and predicted labels. It shows the number of correct predictions as well as misclassifications, helping to evaluate how well the model distinguishes between positive and negative sentiments.

```
review = ["This movie was absolutely amazing and I loved it"]

# transform using same vectorizer
review_vec = vectorizer.transform(review)

# predict
prediction = model.predict(review_vec)

if prediction[0] == 1:
    print("Positive Review 😊")
else:
    print("Negative Review 😞")
```

[46]

```
Positive Review 😊
```

```
review = ["worst movie ever"]

# transform using same vectorizer
review_vec = vectorizer.transform(review)

# predict
prediction = model.predict(review_vec)

if prediction[0] == 1:
    print("Positive Review 😊")
else:
    print("Negative Review 😞")
```

[49]

```
Negative Review 😞
```

Fig 8: Sentiment Prediction on Sample Reviews

This output demonstrates the model predicting sentiment for new text inputs. The model correctly classifies a positive review and a negative review, showing its effectiveness in real-time sentiment prediction.

CONCLUSION

The internship provided valuable exposure to real-world data analysis and machine learning applications. It helped in understanding how raw data can be collected, cleaned, processed, and analyzed to extract meaningful insights. Through various tasks, the importance of data preprocessing, feature selection, and visualization in building accurate models was clearly understood.

The hands-on tasks significantly improved technical skills in Python programming, data visualization, and machine learning techniques. Working with libraries such as Pandas, NumPy, Matplotlib, Scikit-learn, Plotly, and Dash enhanced practical knowledge and

confidence in implementing real-time projects. The development of models and dashboards also strengthened the ability to translate theoretical concepts into practical solutions.
In addition to technical skills, the internship enhanced analytical thinking, problem-solving ability, and logical reasoning. It provided experience in handling real-world datasets, identifying patterns, and making data-driven decisions. The challenges faced during implementation helped in improving debugging skills and understanding model limitations.
Furthermore, the internship improved time management, documentation, and presentation skills. It encouraged independent learning and adaptability to new tools and technologies.
Overall, this internship experience has played a crucial role in preparing for future opportunities in the field of data science and analytics. It has built a strong foundation in data handling, model development, and visualization, making it easier to approach more advanced concepts and real-world projects with confidence.

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7. Géron, A., *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media

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INTERNSHIP OFFER LETTER

CODTECH IT SOLUTIONS PVT.LTD

INFORMATION TECHNOLOGY SERVICES

8-7-7/2, Plot NO.51, Opp: Naveena School, Hasthinapuram Central, Hyderabad , 500 079. Telangana

Internship Offer Letter

To whom so ever it may concern

Candidate Name: Yamuna V

Selected For: Data Analytics

Organization: Codtech IT Solutions Private Limited

Duration: 12 Weeks

Internship Period: 26 February 2026 - 21 May 2026

Intern ID: CTIS6475

This internship with training is designed as an educational experience, focusing on learning, skill development, and gaining practical knowledge. As an intern, we expect you to:

1. Complete all assignments to the best of your ability.
2. Follow any lawful and reasonable instructions provided by your supervisors.
3. Participate actively in team meetings and discussions.
4. Provide regular updates on your progress.
5. Adhere to company policies and maintain a professional demeanor.
6. Collaborate effectively with team members and contribute to group projects.
7. Seek feedback and apply it to improve your performance.

We trust that you will approach all tasks with diligence and enthusiasm. We are confident that this internship will be an enriching experience for you. We look forward to working with you and supporting you in achieving your career aspirations

#startupindia

Best regards,

N. Santhosh

Neela Santhosh Kumar

Human Resources & Academic Head

CODTECH IT SOLUTIONS PRIVATE LIMITED

VERIFIED BY

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